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Enhancing EV Charging Station Security Using A Multi-dimensional Dataset : CICEVSE2024

Emmanuel Dana Buedi

dana.buedi@unb.ca

University of New Brunswick https://orcid.org/0009-0008-5612-0104

Ali A. Ghorbani

University of New Brunswick https://orcid.org/0000-0001-9189-6268

Sajjad Dadkhah

University of New Brunswick https://orcid.org/0000-0002-5582-0255

Raphael Lionel Ferreira

University of New Brunswick

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Enhancing EV Charging Station Security Using A Multi-dimensional Dataset : CICEVSE2024

Emmanuel Dana Buedi[1][0009−0008−5612−0104], Ali A. Ghorbani[1][0000−0001−9189−6268], Sajjad Dadkhah[1][0000−0002−5582−0255], and Raphael Lionel Ferreira[1]

University of New Brunswick, 3 Bailey Dr, Fredericton NB, Canada
{dana.buedi, ghorbani, sdadkhah, raphael.ferreira}@unb.ca

Abstract. The rapid adoption of electric vehicles (EVs) is fundamentally transforming the automotive industry, prompting a surge in the installation of charging stations to accommodate the growing number of EVs and enhance overall mobility and user experience. Efforts to conduct machine learning-based cybersecurity research and developing solutions to address the growing threats and vulnerabilities in EV charging station infrastructure face challenges stemming from the unavailability of suitable datasets. The primary contribution of this study is addressing these challenges by publishing a multi-dimensional dataset that comprises power consumption data, network traffic and host activities of the EVSE in both benign and attack conditions. The experimental testbed utilizes a real EVSE, Raspberry Pi and standard industry communication protocols for EV charging infrastructure, with the scenarios observing the EVSE in both idle and charging states. The results of statistical analysis and machine learning classification tasks demonstrate the suitability of this dataset for baseline behavioral profiling, classification and anomaly detection tasks.

Keywords: Electric Vehicle Charging Station · Multi-dimensional Dataset · Cybersecurity.

1 Introduction

The automotive landscape is undergoing a profound transformation with the rapid proliferation of electric vehicles (EVs). It is anticipated that electric cars will account for more than 60 percent of all vehicles sold globally by the year 2030 [8], indicating a substantial shift from the existing internal combustion engine (ICE) paradigm. This surge is fueled by a confluence of factors, including advances in battery technology, government incentives and the ever-increasing demand for sustainable mobility solutions focused on environmental consciousness [17, 33]. As the number of electric vehicles on the road increases, more charging stations are being installed to facilitate mobility and improve the user experience.

The Electric Vehicle Supply Equipment (EVSE) integrates both hardware and software elements and is designed to facilitate seamless and safe operations,

catering to the diverse needs of electric vehicle users and charging station operators [37]. In order to match or surpass the user experience and convenience provided by conventional fuel stations, the electric vehicle charging infrastructure has been improved through the integration of communication and software features. EV users have the option to select from a variety of authentication and billing services [43]. This includes External Identification Means (EIM) like QR codes, RFID-enabled cards and NFC scanning [36], or a more user-centric approach known as Internal Identification Methods (IIM), where the EV self-authenticates through methods like Autocharge [3] and Plug&Charge [27]. The implementation of IIM requires Vehicle-to-Grid (V2G) communication using ISO 15118 [26] between the EV and EVSE which is achieved using Power Line Communication (PLC) via the HomePlug Green PHY modem through the Control Pilot (CP) interface. The EVSE utilizes the Open Charge Point Protocol (OCPP) [2] to establish remote communication with the Charging Station Management System (CSMS), providing authentication, authorization, and remote monitoring services. The hardware implementation of the EVSE as described in [16] and presented in Figure 2, reflects the level of sophistication and supported services.

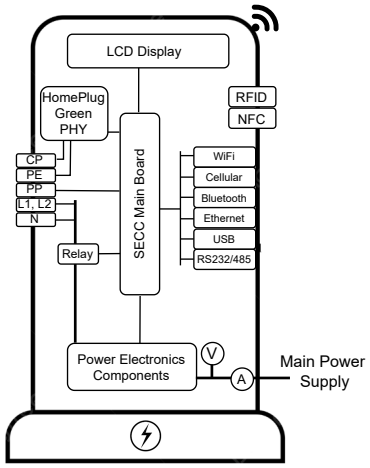


Fig. 1: EVSE Hardware Components

While improving user experience and convenience through enhanced services is desirable, the associated hardware and software features may be susceptible to cybersecurity threats [15]. Researchers are actively addressing the vulnerabilities and threats in electric vehicle charging infrastructure [4, 38, 40, 34, 15]. Machine learning models have emerged as promising tools for cybersecurity applications, particularly in intrusion and anomaly detection. However, these models rely on datasets that comprise the charging station's behavior under both normal and anomalous conditions. Currently, the application of machine learning models

in cybersecurity research for electric vehicle charging stations faces several open issues relating to availability of datasets. The existing datasets primarily concentrate on EV charging sessions and revolve around modeling and planning power system load profiles. Datasets sourced from other domains, such as the Internet of Things (IoT) and Operational Technology (OT), fail to offer a comprehensive representation of the operational behavior of a charging station, rendering them unsuitable as alternatives. There are no publicly available datasets that capture the internal behavior and communication network activities of the EVSE host under both normal and anomalous conditions. The main contribution so this work are:

- Generating and publishing a multi-dimensional dataset constituting power consumption data, network traffic captures and host activities of the EVSE under normal and anomalous conditions
- Performing statistical analysis on the datasets, comparing some features to assess their adequacy for behavioral profiling and anomaly detection tasks
- Evaluating the suitability of the dataset for binary and multi-class classification of benign and attack scenarios using machine learning algorithms

The remainder of this paper is organized as follows: Section 2 compares the contribution of this work to other related research. Section 3 presents the details of the EV charging station Lab setup and the various scenarios implemented. Section 4 presents the dataset and statistical analysis of some key features. Section 5 presents a machine learning evaluation of the dataset for the classification of benign and attack scenarios. Finally, Section 6 presents the conclusion and major findings of this research work.

2 Related Work

Within Cybersecurity research, this work can be positioned in the domain of studying the baseline behaviour of EV charging stations for intrusion or anomaly detection purposes. In this regard, the closely related work, [22] focuses on EV authentication within the charging station infrastructure. In their work, they set up a simulation environment using a Linux-based general-purpose computer. Three main entity categories, Electric vehicles, charging stations and a Grid Servers were simulated on the host computer. The Perf tool is then used to record specific kernel events from the host computer for each entity referenced by the process ID. They considered four main EV authentication DDoS attack scenarios involving a combination of full, random, gaussian and non-gaussian attacks. On the other hand, our work generates network, host and power consumption data from a testbed setup consisting of an actual EV charging Station and Raspberry Pi devices. The attack scenarios in our work include both host and network based attacks.

In the area of characterizing EV charging behaviour based on charging session data, a lot of work has been done to generate benchmark datasets to support ongoing research. Typically, charging session features includes EV power

Table 1: Existing Datasets for Electric Vehicle Charging Stations

Dataset	Year	Hardware Device Platform	Data	EVSE Communication Protocols	Attack Scenario(s)	Accessibility
Pecan Street Dataport [19]	2023	EVSE	Energy Consumption	None	None	Commercial
CICEV2023 [22]	2023	General Purpose Computer	Host Events	None	Network Attacks	Public
Asensio [6]	2020	EVSE	Charging Sessions	None	None	Public
ACN-data [23]	2019	EVSE, Mobile App	Charging Sessions, EV User	None	None	Public
E. Xydas et al [48]	2016	EVSE	EV Charging Demands	None	None	Private
iMove [11]	2015	EVSE	Charging Sessions	None	None	Public
ELaadNL [11]	2015	EVSE	Charging Sessions	None	None	Public
This Work	2024	EVSE, Raspberry Pi	Network, Host, SECC Power Consumption	OCPP, ISO15118 (V2G)	Network and Host Attacks	Public

consumption, EV-to-EVSE connection and disconnection time, EV arrival and departure time, charging current and status of some pilot signals. The Adaptive Charging Network, ACN-Data [23] provides a comprehensive, up-to-date and publicly accessible dataset on EV charging sessions. The network started with over 100 EVSEs located in two sites within the Caltech and JPL campuses. They collect data directly from the charging stations and the EV user mobile application. In their work, [48] monitored the EV charging demands for 255 charging stations and weather data from 2012 to 2013. Similar to the ACN-Data, the Asensio [6], Pecan Street Dataport [19], iMove and ELaadNL [11] all focus on charging session data. The focus of these datasets has been on the overall impact of EV charging station demand on the electrical grid. Even though these works could be used for profiling EV charging sessions and to some extent, EV user behaviours, they however unlike our work, are not cybersecurity focused and hence cannot be used for detecting security attacks launched against the EVSE.

Another associated area of study explores the application of hardware events for cybersecurity solutions, including device behavioral fingerprinting for anomaly

detection [39, 30, 47] and malware classification [31, 41, 1]. However, the datasets used for most of these works are unavailable for further experiments and validation. Distinguishing itself from previous studies, this research introduces additional data from sources such as power consumption and network traffic, providing researchers with a publicly available dataset with a broader range of options for detecting malicious activities in resource constrained devices.

3 EV Charging Station Lab Setup

Our testbed configuration setup presented in Figure 2 comprises an operational Level 2 charging station [42], EVSE-A and a Raspberry Pi and communication setup. The EVSE-A is configured to communicate to the remote CSMS platform using OCPP protocol and could be installed for use in a public, home or shared facility [5] . The EVSE-B is represented by a Raspberry Pi, it communicates with the EVCC and CSMS using ISO15118 and OCPP respectively. The power consumption of the EVSE-B is also monitored by another Raspberry Pi using a wattmeter and I2C protocol [24].

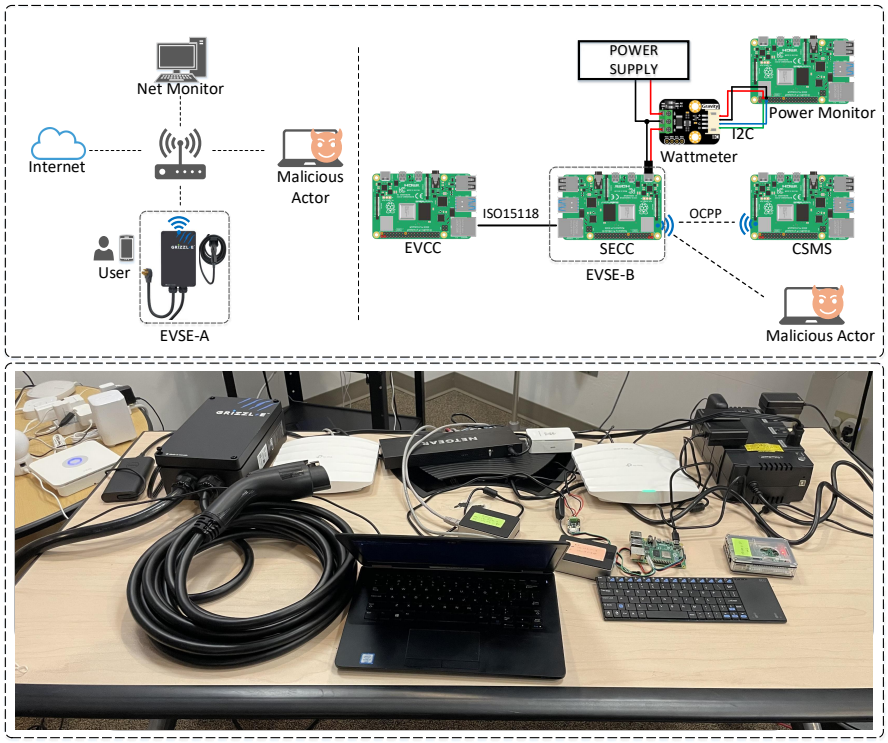


Fig. 2: EV Charging Station Lab Setups

3.1 Electric Vehicle Supply Equipment

The EVSE-A employed in our laboratory setup is the Grizzl-E smart EV charger, a Level 2 AC charging unit supporting a 40A / 9.6kW charging output. The ESP32 controller powers the intelligent functionalities of this EV charger, providing WiFi capabilities for local and remote communication. A detailed description of the EVSE features is provided in Table 2.

Table 2: EVSE Specifications

Feature	Specification
Model	Grizzl-E Smart Connect - GRS-6-24-PB
Type	AC Level 2 (208-240 VAC)
Maximum Output Ratings	40A (9.6 kW)
Charging cable Connector	J1772
Connectivity	2.4Ghz Wifi network
Mobile App	Yes
Protocols	OCPP 1.6
Main board controller	ESP32
Status Indicators	LED, Buzzer

The EVSE-B is represented by a Raspberry Pi 4 model B, featuring a Quad-core Cortex-A72 (ARM v8) 64-bit processor, 2GB RAM, running Ubuntu 22.04.3 LTS (Jammy), and utilizing the Linux 5.15.0-1044-raspi kernel. The deliberate choice of Raspberry Pi hardware is motivated by several key factors that highlight its suitability for this project:

- **Alignment with EVSE Main Board Hardware Requirements:** The features available on Raspberry Pi closely align with the essential requirements of typical EVSE hardware, as illustrated in Figure 2. This strategic alignment ensures that our lab setup accurately mirrors the characteristics of the main board in real-world EVSE, fostering a more representative and effective testing environment.
- **Proven Implementation in Existing EV Charging Stations:** Raspberry Pi hardware has a track record of successful implementation in various commercial EVSEs. Leveraging this established compatibility not only validates the choice but also underscores the practicality and reliability of Raspberry Pi in the context of electric vehicle charging infrastructure. This familiarity contributes to the robustness of our experimental configuration [46, 35, 18].
- **Accessibility and Replicability:** The widespread availability and ease of acquisition of Raspberry Pis enhance the accessibility of our research methodology. This deliberate choice ensures that other cybersecurity researchers can readily replicate our setup for their investigations.

3.2 Power Consumption Circuit

The power consumed by the SECC or main board of EVSE-B is monitored and recorded to serve as additional data used for baseline profiling of the EVSE. To achieve this, we used the I2C Digital Wattmeter [12] sensor board. This board supports load within 26VDC, 8A ratings. The input of the module is connected in series between the power supply adapter and the EVSE-B. The output of the module is connected to the power monitoring Raspberry Pi through the SDA and SCL pins using the I2C protocol. We used a Python script to log the data from the Wattmenter, the script is in two parts:

- **Initialization:** This part of the script initializes the I2C interface of the Raspberry Pi, loads the calibrations for the Wattmeter and creates a CSV file to log the data.
- **Logging Loop:** This section of the script iteratively queries the I2C bus, employing functions such as getBusVoltage_V(), getShuntVoltage_mV(), getCurrent_mA(), and getPower_mW(). The purpose is to retrieve and record data from the Wattmeter, executing this process at a regular sampling interval of 1 second.

3.3 Network Topology

The network packets are captured from the ISO15118 interface of EVSE-B and the OCPP interfaces of EVSE-A and EVSE-B using Wireshark and TCPdump [7] on a Linux computer. The Ethernet connection of the two Access Points, AP1 and AP2 are mirrored [49] to the monitoring and capturing port of the switch as presented in Figure 3. This allows the network capture tool to gain access to all the network communication traffic traversing the switch, facilitating the capture of the specific traffic of interest. Attack traffic generated by the Attacker and benign traffic communication between any of the EVSE and the CSMS or EVCC can be captured with this topology.

3.4 Experiment Scenarios

Our experiments include scenarios that depict both the normal operational patterns and attack conditions of the EVSE. Under normal operations, the EVSE is allowed to carry out its routine functions in both idle and charging states. For EVSE-B, the status of V2G communication with the EVCC determines its operational state. The benign dataset is captured during normal operational modes in scenarios SNI1 and SNC1, providing valuable information for establishing baseline behavioral profiles of the EVSE. A summary of the various scenarios explored in this experiment is presented in Table 3.

The Attack scenarios execute various host and network based attacks on the EVSE in both the idle and charging states. The details of the executed attacks are presented in Table 4. For network attacks, the communication interfaces of the EVSE is exposed to reconnaissance and Denial of Service (DoS) attacks from

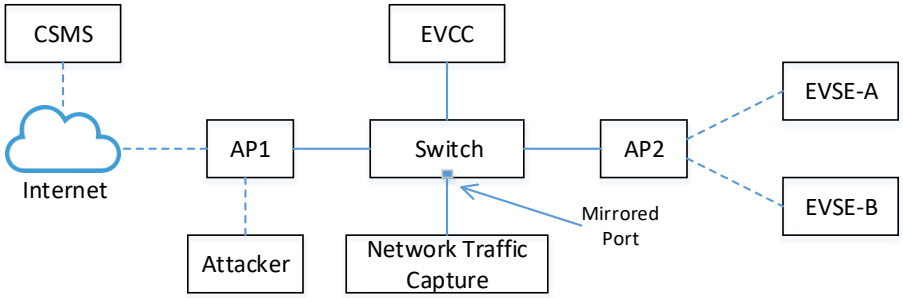


Fig. 3: Topology used for Capturing Network Traffic pcap files. The computer hosting the Wireshark and TCPdump monitoring and capture tools is connected to the mirrored port of the Ethernet switch.

Table 3: Experiment Scenarios

Scenario	EVSE State	Behavior	Description
SNI1	Idle	Benign	– No connection to EVCC – OCPP heartbeat communication between EVSE and CSMS
SNC1	Charging	Benign	– EVCC Connected to EVSE-B with ISO 15118 V2G Communication – OCPP charging data transmission between EVSE and CSMS
SMI1	Idle	Network Attacks	– No conecttion to EVCC – OCPP heartbeat communication between EVSE-A / EVSE-B and CSMS – Scenario for each network attack type describe in Table 4
SMC1	Charging	Network Attacks	– EVCC Connected to EVSE-B with ISO 15118 V2G Communication – OCPP charging data transmission between EVSE-B and CSMS – Scenario for each network attack type describe in Table 4
SMI2	Idle	Host Attacks	– No connection to EVCC – OCPP heartbeat communication between EVSE-B and CSMS – Scenario for each host-based attack type describe in Table 4
SMC2	Charging	Host Attacks	– EVCC Connected to EVSE-B with ISO 15118 V2G Communication – OCPP charging data transmission between EVSE-B and CSMS – Scenario for each host-based attack type describe in Table 4

a malicious actor with access to the wireless interface of the EVSE. We also implement the case where an already compromised EV launches an attack against the V2G interface of the EVSE. A malicious device equipped with a HomePlug Green PHY modem, legitimate MAC address or digital certificate could act as an EV and pose a threat to any V2G communication enabled EVSE. [14]. Executing various scans during the reconnaissance stage provides useful information on open ports, available services and system information. The malicious actor

can proceed to exploit any discovered vulnerabilities to gain privileged access to the EVSE.

For the case of EVSE host based attacks, we assume that the worst-case scenario where the EVSE main board has already been compromised through a discovered vulnerability or through privileged access to the diagnosis interface ports of the EVSE [20]. The attacker performs various malicious activities through an established backdoor connection to a Command and Control (C2) server. The backdoor enables the attacker to execute malicious scripts to perform activities such as

- Payload download
- File access permission changes
- File creation, deletion and checking present working directory
- File encryption and decryption

Another host attack scenario is cryptojacking which refers to the case where an attacker hijacks the computing resource of a victim to mine cryptocurrency [44]. Resource-constrained devices such as IoT are prone to this attack where attackers hijack these devices to download crypto mining malware. Attacker could deploy crypto malware payloads such as Mal/Miner-c [32] to victims devices in a botnet to harvest computing resources into a pool to mine cryptocurrencies. The main controller board or SECC of the EVSE presents a potential target for such an attack due to its constant connection to a power source. Moreover, the auxiliary energy usage by the internal components of the EVSE is often inadequately monitored. In our case, we configure EVSE-B to join a legitimate pool server to mine the Monero[1] cryptocurrency.

4 The CICEVSE2024 Dataset

4.1 Power Consumption Data

The energy consumption data provided by the CSMS and mobile application only indicates the amount of power delivered to the EV during charging [29, 16, 28]. However, the energy consumed by the internal components of the EVSE, such as the main-board is usually not reported. As observed in Figure 4, this data provides a good source of information and to some extent, an indication of the workload of the main-board of the EVSE. Notably, unlike backdoor and Recon, the current drawn during DoS and cryptojacking attack scenarios are significantly higher, reaching almost double the levels observed in benign conditions in certain cases. Depending on the attack, the percentage deviations of current consumed by the main board of the EVSE present a good Indication of Compromise (IOC) useful for building anomaly detection systems for EV charging stations. By analysing the power consumption data we can determine if the workload deviates from the expected baseline behavior.

[1] https://www.getmonero.org/downloads/

Table 4: Host and Network Attacks

Attack					Impact		
Attack Type	Attack Tool	Interface / Target	Description	MITRE ATT&CK Tactics and Techniques	C	I	A
TCP Port Scan	nmap	EVSE-A: OCPP EVSE-B: OCPP & ISO15118	Identify open TCP ports on a target system.	Discovery - Service Discovery (T1007) Network Service Discovery (T1046)	•	○	○
Service Version Detection	nmap	EVSE-A: OCPP EVSE-B: OCPP & ISO15118	Determine the software and version running on open ports.	Discovery - Service Discovery (T1007), Software Discovery (T1518)	•	○	○
OS Fingerprinting	nmap	EVSE-A: OCPP EVSE-B: OCPP & ISO15118	Attempt to identify the operating system.	Reconnaissance - Gather Victim Host Information (T1595) Discovery - System Information Discovery (T1082)	•	○	○
Aggressive Scan	nmap	EVSE-A: OCPP EVSE-B: OCPP & ISO15118	Combine various scan types for a comprehensive view of the target.	Reconnaissance - Gather Victim Host Information (T1595) Discovery - Service Discovery (T1007), Collection (T1119)	•	○	○
SYN Stealth Scan	nmap	EVSE-A: OCPP EVSE-B: OCPP & ISO15118	Use SYN packets to identify open ports.	Discovery - Service Discovery (T1007)	•	○	○
Vulnerability Scan	nmap	EVSE-A: OCPP EVSE-B: OCPP & ISO15118	Systematically scanning network interface to identify potentially exploitable security flaws.	Reconnaissance - Gather Victim Host Information (T1595) Discovery - System Information Discovery (T1082) Network Service Discovery (T1046)	•	○	○
Slowloris Scan	nmap	EVSE-A: OCPP	Explot local webserver on EVSE by keeping multiple connections, exploit limitations on concurrent connections to exhaust its resources and render it unresponsive.	Impact - Endpoint Denial of Service (T1499)	○	○	•
UDP Flood	hping3	EVSE-A: OCPP EVSE-B: OCPP & ISO15118	Sends large volumnes of UDP packets to overwhelm targeted network interfaces.	Impact - Network Denial of Service (T1498), Endpoint Denial of Service (T1499).	○	○	•
ICMP Flood	hping3	EVSE-A: OCPP EVSE-B: OCPP & ISO15118	Sends large volumnes of ICMP packets to overwhelm targeted network interfaces.	Impact - Network Denial of Service (T1498), Endpoint Denial of Service (T1499)	○	○	•
PSHACK Flood	hping3	EVSE-A: OCPP EVSE-B: OCPP & ISO15118	Flood attack exploiting the TCP PSH flag to overwhelm targeted network interfaces.	Impact - Network Denial of Service (T1498), Endpoint Denial of Service (T1499)	○	○	•
ICMP Fragmentation	hping3	EVSE-A: OCPP EVSE-B: OCPP & ISO15118	Sending fragmented ICMP packets to exploit vulnerabilities in handling fragmented traffic on the target network interface.	Impact - Network Denial of Service (T1498), Endpoint Denial of Service (T1499)	○	○	•
TCP Flood	hping3	EVSE-A: OCPP EVSE-B: OCPP & ISO15118	Overloading target's network interface with a large volume of TCP packets to disrupt services and cause unresponsiveness.	Impact - Network Denial of Service (T1498), Endpoint Denial of Service (T1499)	○	○	•
SYN Flood	hping3	EVSE-A: OCPP EVSE-B: OCPP & ISO15118	Flooding a target's network interface with a high volume of TCP SYN packets, exploiting the three-way handshake process and causing service disruption.	Impact - Network Denial of Service (T1498), Endpoint Denial of Service (T1499)	○	○	•
SynonymousIP Flood	hping3	EVSE-A: OCPP EVSE-B: OCPP & ISO15118	Sending a flood of packets from synonymous IP addresses, potentially evading detection and overwhelming target network interface.	Impact - Network Denial of Service (T1498), Endpoint Denial of Service (T1499)	○	○	•
Cryptojacking	Monero	EVSE-B	Illegitimate use of victim's computing resources to mine cryptocurrency without consent	Impact - Resource Hijacking (T1496)	•	•	•
Backdoor	C2 Server	EVSE-B	Unauthorised remote access to a victim. For our experiment, the C2 server drops and executes malicious scripts on the victim EVSE. Malicious Activities: Payload download, File access permission changes, File encryption and decryption, File creation, deletion and checking PWD.	Execution - Command and Scripting Interpreter (T1059) Command and Control - Content Injection (T1659) Defense Evasion - File and Directory Permissions Modification (T1222), Indicator Remover (T1070) Impact - Data Encrypted for Impact (T1486) Discovery - File and Directory Discovery (T1083)	•	•	•

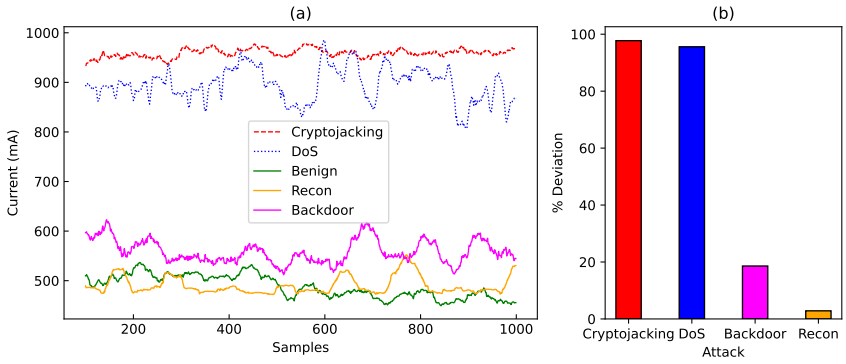


Fig. 4: Wattmeter Current Reading for EVSE-B. (a) presents current reading with MA window size= 20 while (b) shows the percentage deviation of attacks from benign samples.

4.2 HPC and Kernel Events

The EVSE employs resource-constrained or embedded computing hardware as its main board or SECC to execute essential intelligent functions necessary for charging station operation [16]. Conventional host security solutions, such as Anti-Virus software may not be applicable for such resource constrained devices. As an alternative, a viable solution leverages the computing events of the system hardware to establish robust security measures [31]. Modern processors from manufactures such as intel [45], ARM [25] and AMD [13] are equipped with Hardware Performance Counters (HPC), which record low-level microarchitecture events [10].

Table 5: Labelling of the Host Dataset

Column ID	Entry
State	Idle, Charging
Scenario	Recon, DoS, Cryptojacking, Backdoor, Benign
Attack	Cryptojacking, Backdoor, None, TCP-port-scan, service-version-detection, os-fingerpriting, aggressive-scan, syn-stealth-scan, vulnerability-scan, slowloris-scan, upd-flood, icmp-flood, pshack-flood, icmp-fragmentation, TCP-flood, syn-flood, synonymousIP-flood
Label	Attack, Benign
Interface	OCPP, ISO15118

These hardware events prove instrumental in baseline behavioral fingerprinting of devices, classifying malware, and employing machine learning techniques

for anomaly detection [9]. We utilize PERF[2] to gather approximately 900 kernel and HPC events from the Raspberry Pi (EVSE-B) during the experimental scenarios.

Table 5 provides a summary of the host dataset labels for each scenario. Figure 5 illustrates a T-SNE visualization of the samples, showcasing the grouping of samples according to the charging station's state and the various attack scenarios. Notably, T-SNE exhibits its capability to effectively represent this high-dimensional dataset in two dimensions, revealing a clear separation between attack and benign scenarios.

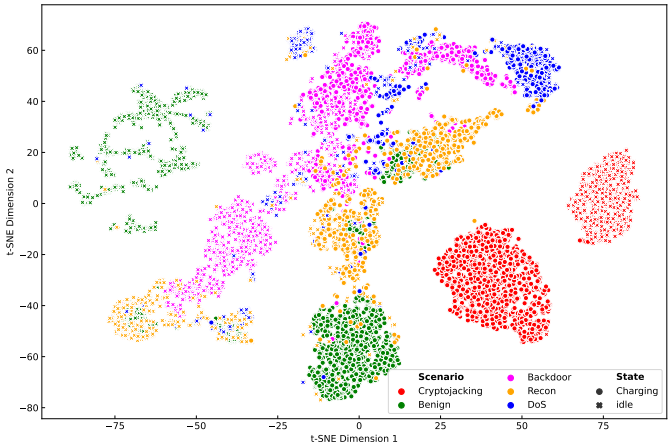


Fig. 5: T-SNE visualisation of Hardware Events Samples Grouped by Scenarios and Charging State of EVSE

Figure 6 depicts a plot illustrating two HPC events and two kernel events recorded for the experiment scenarios. In the case of host-based attacks such as Cryptojacking, there is a notable increase in instruction execution and memory access, reaching up to 800% of the benign scenario. A comparable pattern is observed for network-based attacks. However, reconnaissance attacks exhibit a comparatively lower deviation from the benign scenario, making them potentially challenging to detect using HPC or kernel events.

In Table 6, the mean deviations of some HPC and kernel events have been computed. Generally, the malicious scenarios is observed to exhibit a substantial deviation from benign samples. These events offer valuable insights for crafting lightweight anomaly detection security solutions tailored to resource-constrained devices, such as the EVSE in this context. Rule-based algorithms can be leveraged to formulate IF-THEN rules, providing a foundation for identifying malicious system behavior.

[2] https://perf.wiki.kernel.org/index.php/Main_Page

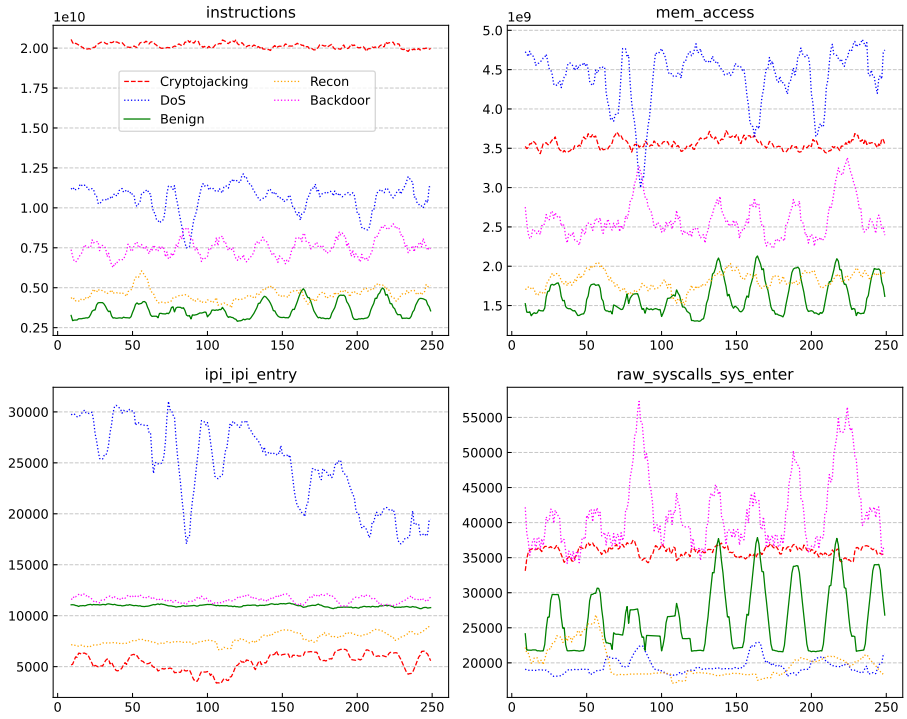


Fig. 6: HPC and Kernel Events under Benign and Attack Conditions.
MA window size = 10

5 Machine Learning Experiments

We assess the dataset's suitability for machine learning applications by conducting multi-class and binary classification tasks using eight machine learning algorithms. The network traffic, HPC and kernel event datasets are evaluated independently to facilitate a comparison of algorithm performance on each of these data sources. To mitigate the curse of dimensionality, we employ Principal Component Analysis (PCA) [21] for feature extraction during the pre-processing stage. Subsequently, we utilize 10-fold cross-validation during the model training and testing stages in an attempt at generalization and to mitigate the impact of model over-fitting. In the binary classification task, the models are trained to classify samples as either benign or attack using these data sources separately. In the multi-class classification task, we employ *Scenario* labels with five classes — Cryptojacking, Backdoor, Recon, DoS, and Benign as presented in Table 5.

The results of the machine learning classification experiments are presented in Figure 8. Across both binary and multi-class classification tasks, the HPC data source consistently outperforms kernel events in nearly all experiment instances. Notably, the models exhibit superior performance in the binary classification

Table 6: Selected HPC and Kernel Events : Computed Average Percentage Deviations of Attack Sampled from Benign Samples

Event	Description	Backdoor	Crypto	Dos	Recon
instructions	Number of instructions executed	214.19	943.54	316.65	130.69
cache-misses	Number of cache misses	39.92	397.06	112.83	64.70
exc_taken	Exception taken	86.82	61.42	169.62	44.55
cpu-migrations	CPU Migrations	842.43	-100.00	1425.70	110.05
dTLB-store-misses	Data TLB - Write Misses	39.56	1068.57	101.48	72.97
l1d_cache_wr	Level 1 data cache access - Write	143.87	121.56	321.24	113.14
L1-icache-loads	Level 1 instruction cache access - Read	163.10	538.63	283.92	107.18
l2d_cache_rd	Level 2 data cache - Read	51.47	422.95	151.50	70.75
mem_access_rd	Data memory access - read	118.19	387.56	253.17	91.42
mem_access_wr	Data memory access - write	142.94	119.96	322.17	114.86
kmem_kfree	Kernel memory freeing event	149.06	67.66	2087.07	187.25
net_dev_xmit	Network device transmission event	27.42	33.79	172.88	59.64
qdisc_dequeue	Dequeue event	31.47	42.66	261.16	88.42
raw_syscalls_sys_enter	System call entry (raw) event	107.66	79.70	2.35	44.13
irq_softirq_raise	Software interrupt - Raised	90.29	65.72	1573.65	109.45
sched_migrate_task	Task migration event in the scheduler	670.49	-100.00	1091.74	78.18
sched_switch	Task switch event in the scheduler	42.10	39.22	332.82	24.59
syscalls_sys_enter_close	System call entry for close syscall	421.15	273.72	15.58	231.63
syscalls_sys_enter_read	System call entry for read syscall	44.32	30.19	37.09	56.21
syscalls_sys_enter_write	System call entry for write syscall	119.30	35.73	57.34	76.92

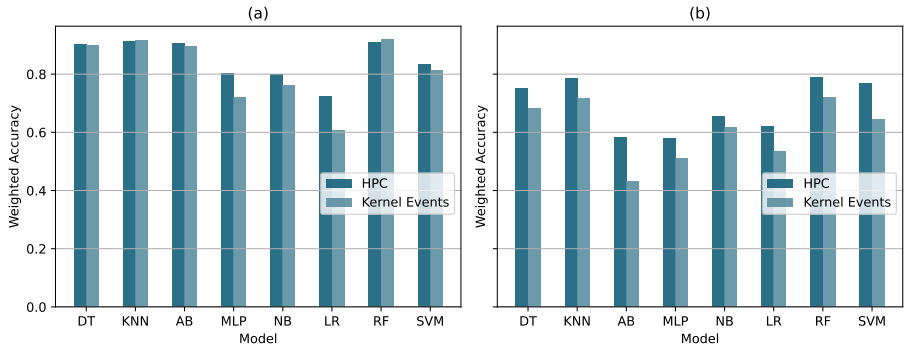


Fig. 7: Weighted Accuracy for HPC and Kernel Events using Different Machine Learning models. The performance of the models for Binary Classification Task is presented in (a) whilst the results for multi-class classification task is presented in (b)

task compared to the multi-class classification task. In the binary classification scenario, Decision Tree, KNN, Adaboost, Random Forest, and Support Vector Machine achieved weighted accuracy exceeding 80%. Tables 7 and 8 provide a comprehensive report on the experiments, presenting details such as weighted accuracy, F1-score, Precision, and Recall for all the models. For binary classification, KNN and Random Forest reported the highest accuracy of more than 91% for HPC and kernel events respectively. For multi-class classification, Random Forest reported the highest accuracy of 78.87% and 71.99% for HPC and

Table 7: Binary Classification Task: Weighted Average Results for HPC and Kernel Events Employing Various Machine Learning Models

Model	HPC				Kernel Events			
	Accuracy	F1-score	Precision	Recall	Accuracy	F1-score	Precision	Recall
Decision Tree	0.901 49	0.901 87	0.902 33	0.901 51	0.900 32	0.900 86	0.901 60	0.900 33
KNN	**0.91342**	**0.91360**	**0.91380**	**0.91343**	0.915 80	0.915 43	0.915 18	0.915 80
Adaboost	0.906 33	0.904 52	0.904 86	0.906 35	0.893 71	0.891 43	0.891 67	0.893 71
MLP	0.801 01	0.804 94	0.811 44	0.801 01	0.720 70	0.723 18	0.726 02	0.720 71
Naive Bayes	0.798 04	0.807 80	0.846 05	0.798 06	0.760 14	0.773 01	0.832 13	0.760 15
Logistic Regression	0.721 88	0.737 09	0.831 29	0.721 89	0.607 20	0.618 65	0.825 77	0.607 22
Random Forest	0.910 70	0.910 60	0.910 49	0.910 72	**0.91887**	**0.91786**	**0.91779**	**0.91887**
SVM	0.833 59	0.833 86	0.834 13	0.833 60	0.813 19	0.793 94	0.808 37	0.813 17

Table 8: Multi-class Classification Task: Weighted Average Results for HPC and Kernel Events Employing Various Machine Learning Models

Model	HPC				Kernel Events			
	Accuracy	F1-score	Precision	Recall	Accuracy	F1-score	Precision	Recall
Decision Tree	0.749 29	0.750 03	0.750 82	0.749 29	0.682 10	0.681 89	0.681 84	0.682 09
KNN	0.785 42	0.777 32	0.773 91	0.785 42	0.716 70	0.711 32	0.714 09	0.716 69
Adaboost	0.583 26	0.569 53	0.568 94	0.583 25	0.432 94	0.442 17	0.502 41	0.432 92
MLP	0.580 06	0.556 10	0.552 70	0.580 06	0.511 81	0.492 55	0.490 84	0.511 80
Naive Bayes	0.654 92	0.608 28	0.573 33	0.654 93	0.617 37	0.577 63	0.554 72	0.617 38
Logistic Regression	0.620 91	0.573 01	0.539 28	0.620 92	0.534 01	0.406 62	0.343 36	0.534 01
Random Forest	**0.78873**	**0.78223**	**0.77831**	**0.78873**	**0.71988**	**0.71732**	**0.71836**	**0.71988**
SVM	0.769 24	0.714 09	0.819 20	0.769 24	0.642 88	0.598 58	0.654 83	0.642 89

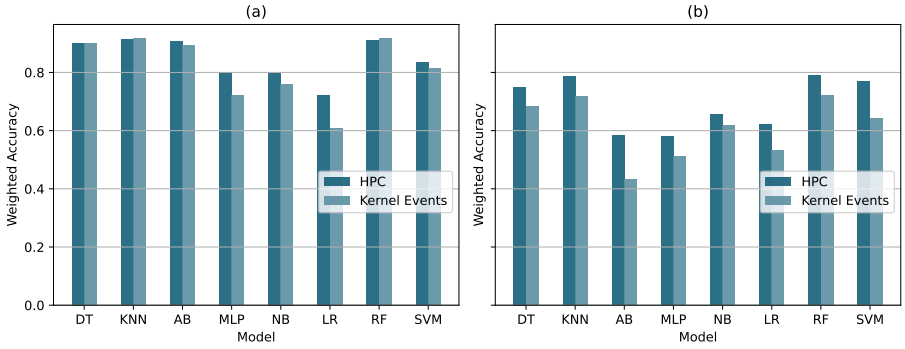
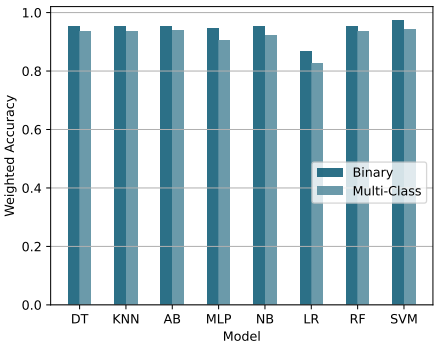


Fig. 8: Weighted Accuracy for HPC and Kernel Events using Different Machine Learning models. The performance of the models for Binary Classification Task is presented in (a) whilst the results for multi-class classification task is presented in (b)

kernel events respectively. The classification results for network traffic data is presented in Figure 9. For Multi-class classification results, the Support Vector Machine (SVM) reported the highest accuracy and recall, with both values reach-



Multi-class scores

Model	Accuracy	F1-score	Precision	Recall
Decision Tree	0.937330	0.947737	0.969380	0.937328
KNN	0.936836	0.947243	0.968804	0.936834
Adaboost	0.937621	0.948028	0.969720	0.937619
MLP	0.905646	0.914749	0.932315	0.905644
Naive Bayes	0.921307	0.930976	0.950198	0.921305
Logistic Regression	0.826176	0.872742	0.942963	0.826176
Random Forest	0.937359	**0.947766**	**0.969414**	0.937358
SVM	**0.941258**	0.927981	0.915085	**0.941258**

Fig. 9: Weighted Scores for Network Traffic Dataset Binary and Multi-class Classification using Different Machine Learning models.

ing 94.13%. On the other hand, Random Forest achieves the highest F1-score at 94.78% and Precision at 96.94%.

6 Conclusion

Investigating cybersecurity solutions for EV charging stations is hindered by the lack of a suitable dataset. This work tackles this pivotal challenge by generating a multi-dimensional dataset that captures network traffic, hardware events, and power consumption data of the EV charging station in both normal and anomalous conditions. Our testbed incorporates a real EV charger, Raspberry Pi, and standard industry communication protocols, enabling us to investigate a variety of attack scenarios. We perform 15 network-based attack scenarios, which include variants of reconnaissance and Denial of Service (DoS), as well as host-based attacks featuring backdoor and cryptojacking. To assess the dataset's suitability for cybersecurity research, we conduct statistical analysis and also, evaluate the use of various machine learning models for classification tasks. We observe a significant variation in the current consumption by the main board of EVSE under attack conditions with reconnaissance exhibiting a notable deviation, while cryptojacking shows the highest level of variation. The outcomes of machine learning classification tasks applied to network traffic and host data demonstrate the capability to distinguish between benign activities and various attack scenarios. The dataset is available publicly for download on the CIC Dataset website[3]. This study lays the groundwork for future research endeavors aimed at developing robust and effective security solutions for electric vehicle charging infrastructure.

[3] https://www.unb.ca/cic/datasets/evse-dataset-2024.html

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